26/08/2026

CODE-C
RM(P3)

Aakash
Medical | IIT-JEE | Foundations

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FORTNIGHTLY TEST FOR NEET-2027 (XII PASSED)

MM: 720 **TE-02C** **Time: 180 Mins.**

Answers

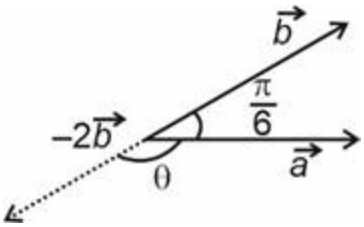
1. (4)	37. (1)	73. (2)	109. (3)	145. (1)
2. (4)	38. (4)	74. (1)	110. (2)	146. (2)
3. (1)	39. (3)	75. (4)	111. (1)	147. (2)
4. (2)	40. (2)	76. (2)	112. (4)	148. (3)
5. (2)	41. (3)	77. (4)	113. (2)	149. (1)
6. (1)	42. (2)	78. (3)	114. (3)	150. (1)
7. (2)	43. (2)	79. (1)	115. (2)	151. (3)
8. (2)	44. (1)	80. (3)	116. (1)	152. (2)
9. (2)	45. (3)	81. (4)	117. (4)	153. (2)
10. (4)	46. (3)	82. (4)	118. (4)	154. (1)
11. (3)	47. (3)	83. (3)	119. (2)	155. (3)
12. (2)	48. (1)	84. (4)	120. (4)	156. (3)
13. (1)	49. (2)	85. (3)	121. (2)	157. (4)
14. (2)	50. (2)	86. (1)	122. (2)	158. (1)
15. (2)	51. (3)	87. (2)	123. (1)	159. (2)
16. (3)	52. (1)	88. (1)	124. (4)	160. (1)
17. (1)	53. (3)	89. (2)	125. (2)	161. (4)
18. (1)	54. (1)	90. (4)	126. (1)	162. (1)
19. (2)	55. (3)	91. (3)	127. (3)	163. (2)
20. (4)	56. (3)	92. (3)	128. (3)	164. (4)
21. (2)	57. (2)	93. (1)	129. (3)	165. (4)
22. (4)	58. (4)	94. (2)	130. (3)	166. (3)
23. (2)	59. (4)	95. (3)	131. (3)	167. (1)
24. (2)	60. (4)	96. (3)	132. (3)	168. (2)
25. (1)	61. (1)	97. (2)	133. (2)	169. (3)
26. (1)	62. (1)	98. (2)	134. (4)	170. (3)
27. (1)	63. (3)	99. (2)	135. (3)	171. (3)
28. (3)	64. (3)	100. (3)	136. (3)	172. (1)
29. (3)	65. (2)	101. (3)	137. (2)	173. (4)
30. (1)	66. (1)	102. (2)	138. (2)	174. (1)
31. (2)	67. (1)	103. (2)	139. (2)	175. (4)
32. (3)	68. (1)	104. (3)	140. (3)	176. (2)
33. (1)	69. (4)	105. (4)	141. (3)	177. (3)
34. (1)	70. (4)	106. (2)	142. (2)	178. (2)
35. (4)	71. (2)	107. (3)	143. (1)	179. (4)
36. (3)	72. (4)	108. (3)	144. (2)	180. (2)

Hints and Solutions

PHYSICS

(1) **Answer :** (4)
Solution:
A vector is not changed if it slides parallel to itself.

(2) **Answer :** (4)
Solution:



$\theta = \pi - \frac{\pi}{6}$
$= \frac{5\pi}{6}$

(3) **Answer :** (1)
Solution:

Only statement I is correct while statement II is incorrect because $\left|\lambda \vec{A}\right| \neq \lambda\left|\vec{A}\right|$ if λ is a negative number.

(4) **Answer :** (2)
Solution:
$\tan\alpha = \frac{5}{5} = 1 \Rightarrow \alpha = \frac{\pi}{4}$

(5) **Answer :** (2)
Solution:
$\cos\alpha,\ \cos\beta$ and $\cos\gamma$ are direction cosines.
$\cos^2\alpha + \cos^2\beta + \cos^2\gamma = 1$

(6) **Answer :** (1)
Solution:
$\vec{v} = \vec{u} + \vec{a}t$
$= \left(4\hat{i} + 6\hat{j}\right) + \left(\hat{i} + \hat{j}\right) \times 2$
$= \left(6\hat{i} + 8\hat{j}\right)$ m/s

(7) **Answer :** (2)
Solution:
$v_x = \frac{dx}{dt} = 6t^2$
$V_y = \frac{dy}{dt} = 8t$
$\vec{v} = 6t^2\hat{i} + 8t\hat{j}$
$t = 1\,\text{sec},\ \vec{v} = 6\hat{i} + 8\hat{j}$
$\Rightarrow \left|\vec{v}\right| = 10\ \text{m/s}$

(8) **Answer :** (2)
Solution:
Relative acceleration
$\vec{a}_{12} = \vec{a}_1 - \vec{a}_2$
= 0
Hence, path will be a straight line

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(9) **Answer :** (2)
Solution:
$t_{\min} = \frac{w}{u} = \frac{200}{4} = 50\ \text{s}$

(10) **Answer :** (4)
Solution:
At the top of trajectory of a projectile acceleration is *g*.

(11) **Answer :** (3)
Solution:
In oblique projectile motion, speed first decreases and then increases while acceleration remains constant.

(12) **Answer :** (2)
Solution:
$v = \sqrt{2}v\cos\theta$
$\cos\theta = \frac{1}{\sqrt{2}}$
$\theta = 45^\circ$

(13) **Answer :** (1)
Solution:
$y = a \text{ x } \frac{-b}{2}x^2$
tanθ = *a*, *b* = $\frac{g}{u^2\cos^2\theta}$
So, $\boxed{R = \frac{2a}{b}}$

(14) **Answer :** (2)
Solution:
At same vertical position, magnitude of velocity remains same

(15) **Answer :** (2)
Solution:
$20\sqrt{2}\cos 60^\circ = v\cos 45^\circ$
$20\sqrt{2}\times\frac{1}{2} = \frac{v}{\sqrt{2}}$
v = 20 m s^{-1}

(16) **Answer :** (3)
Solution:
$\vec{u}_A = \hat{i}+\hat{j} \quad \vec{u}_B = \hat{i} \quad \vec{u}_C = \hat{i}-\hat{j}$
$u_A = \sqrt{2}, \quad u_B = 1, \quad u_C = \sqrt{2}$
$\frac{u_B}{u_C} = \frac{1}{\sqrt{2}}$

(17) **Answer :** (1)
Solution:
$T = \frac{2u_y}{g} = \frac{2\times 8}{10} = \frac{8}{5}\text{s}$

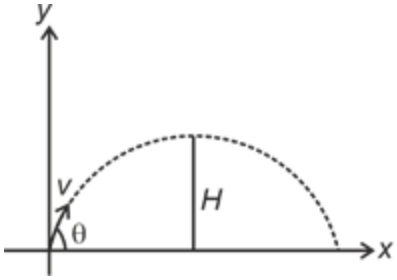
(18) **Answer :** (1)
Solution:
$a = \sqrt{(\alpha r)^2 + (r\omega^2)^2} = r\sqrt{\alpha^2+\omega^4}$
At t = 1 s
ω = 2 rad s^{-1}, α = 2t + 1 = 3 rad s^{-2}
$a = 1\sqrt{3^2+2^4} = 5\ \text{ms}^{-2}$

(19) **Answer :** (2)
Solution:
$x = 5t \Rightarrow t = \frac{x}{5}$
$y = 4t - \frac{1}{2}\times gt^2$
$y = 4t - 5t^2$
$y = \frac{4}{5}x - \frac{x^2}{5}$

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(20) **Answer :** (4)
Solution:
The angle is 90° for uniform circular motion only. For circular motions with increasing and decreasing speeds, the angle is acute and obtuse respectively.

(21) **Answer :** (2)
Solution:



Maximum height reached by the ball

$H = \frac{v^2 \sin^2\theta}{2g}$

$= \frac{(20)^2 \sin^2 30^\circ}{2g}$

$H = 5$ m

(22) **Answer :** (4)
Solution:

$\Rightarrow \ \vec{v} = 3y^2\hat{i} + 2x\hat{j} \ \Rightarrow \ \frac{dx}{dt} = 3y^2$ ...(i)

$\frac{dy}{dt} = 2x$ ...(ii)

$\frac{dx}{dy} = \frac{3}{2}\frac{y^2}{x}$

$\Rightarrow \ 2x\,dx = 3y^2 dy \ \Rightarrow x^2 - y^3 =$ constant

(23) **Answer :** (2)
Hint:

$\vec{F} = m\vec{a}$

Solution:

$m = \frac{\left|\vec{F}\right|}{\left|\vec{a}\right|}$

$= \frac{\sqrt{4+9}}{2\sqrt{2}}$

$= \sqrt{\frac{13}{8}}$ kg

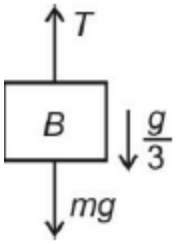
(24) **Answer :** (2)
Hint:
Pulley – mass systems
Solution:
acceleration (a) of blocks

$= \left\{\frac{(m_A+m_B)-m_C}{(m_A+m_B)+m_C}\right\}g$

$= \left\{\frac{8-4}{8+4}\right\}g$

$a = \frac{4}{12}g = \frac{g}{3}$

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$mg - T = \frac{mg}{3}$

$T = \frac{2mg}{3} = \frac{2\times2\times10}{3} = \frac{40}{3}$ N

(25) **Answer :** (1)
Hint:
v = *u* + *at*
F = *ma*
Solution:
m = 5 kg, *t* = 8 s
After the force stops acting the body moves with uniform velocity

$v = \frac{\text{Distance covered}}{\text{Time taken}} = \frac{80}{5} = 16$ m/s

v = *u* + *at*
16 = 0 + *a* × 8
a = 2 m/s^2
F = *ma* = 5 × 2 = 10 N.

(26) **Answer :** (1)
Hint:

Force, $\vec{F} = \frac{d\vec{P}}{dt}$

Solution:

$\frac{dP}{dt} = (2t^2 + 4t + 1)$

$\int_0^P dP = \int_0^t (2t^2 + 4t + 1)dt$

$P = \left[\frac{2t^3}{3} + \frac{4t^2}{2} + t\right]_0^1$

$= \frac{2}{3} + 2 + 1$

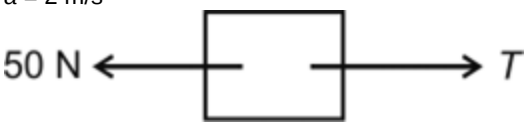
= $\frac{11}{3}$ kg m/s

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(27) **Answer :** (1)
Solution:

$a = \frac{F}{m_1+m_2} = \frac{16}{8} = 2$ m/s^2

(28) **Answer :** (3)
Solution:
The tension in the spring will be the reading of dynamometer
F_{ext} = *Ma*
50 – 30 = 10(*a*)
a = 2 m/s^2



for 6 kg block,
50 –*T* = 6(2)
T = 38 N

(29) **Answer :** (3)
Solution:

$v = \frac{0.1}{2} \times (200) = 10$ m/s

(30) **Answer :** (1)
Solution:
$T = (m_1 + m_2)g = (5 + 10)g = 150$ N

(31) **Answer :** (2)
Hint:
Pseudo-force on the block $F_P = ma$
Solution:
$F_P = ma$
$= 2 \times 4$
$= 8$ N
$[f_s]_{\max} = \mu_s N$
$= 0.5 \times 2 \times 10$
$= 10$ N
$F < [f_s]_{\max}$
$f = F$
$\Rightarrow f = 8$ N

(32) **Answer :** (3)
Hint:
Coefficient of static friction,
μ = tanθ
Solution:
$\frac{1}{\sqrt{3}} = \tan\theta$
$\theta = \tan^{-1}\left(\frac{1}{\sqrt{3}}\right)$
θ = 30°

(33) **Answer :** (1)
Solution:
$2 \times 60\mu = 30$ N
$\mu = \frac{1}{4}$
$= 0.25$

(34) **Answer :** (1)
Solution:
$v = \sqrt{\mu_s Rg} = \sqrt{0.5 \times 13 \times 10}$
$= \sqrt{65} = 8.06 \approx 8\ \mathrm{m/s}$

(35) **Answer :** (4)
Solution:
If a body is in equilibrium under the action of three forces, then their resultant is zero and they form a closed triangle, so they are coplanar.

(36) **Answer :** (3)
Solution:
Pseudo force is associated with non-inertial frame.
A frame moving with uniform velocity is an inertial frame of reference.

(37) **Answer :** (1)
Solution:

A. Average speed = $\frac{\text{Total distance}}{\text{Total time}} = \frac{\frac{2\pi r}{4}}{\frac{2\pi r}{4v}} = v$

B. $|\text{Average velocity}|$

$= \frac{\text{Total displacement}}{\text{Total time}}$

$= \frac{\sqrt{2}r}{\frac{2\pi r}{4v}} = \frac{2\sqrt{2}v}{\pi}$

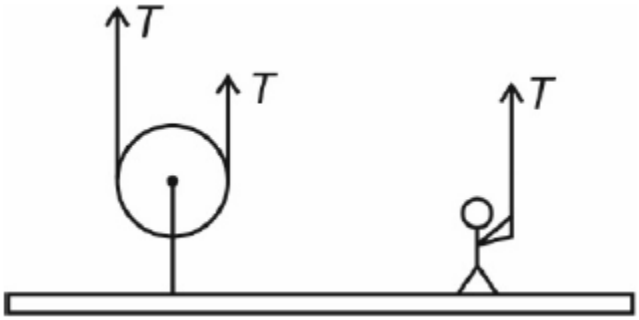
C. |Change in velocity|

$= \left|\vec{v}_B - \vec{v}_A\right| = \sqrt{v_B^2 + v_A^2 - 0} = \sqrt{2}\,v$

D. $\left|\vec{v}_B\right| - \left|\vec{v}_A\right| = v - v =$ zero

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(38) **Answer :** (4)
Hint:
Net force on man plus platform system will be zero.
Solution:
Let force applied by man be *F*
∴ Tension = *T*



3*T* = (25 + 50) g
T = 25 × 10 = 250 N

(39) **Answer :** (3)
Solution:



mg – *T* = *ma*
$mg - \frac{3}{4}mg = ma$
$\frac{g}{4} = a$

(40) **Answer :** (2)
Solution:
$\tan\theta = \frac{v^2}{rg} = \frac{10\times10}{10\sqrt{3}\times10} = \frac{1}{\sqrt{3}}$
⇒ θ = 30°

(41) **Answer :** (3)
Solution:
$\tan 30^\circ = \frac{a}{g} \Rightarrow a = \frac{g}{\sqrt{3}}$

(42) **Answer :** (2)
Solution:
Change in momentum = *m* Δ *v*
Given *m* = 0.15 kg
Δ*P* = 0.15 ×12 – (– 0.15 × 12) = 3.6 Ns

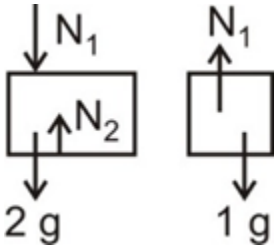
(43) **Answer :** (2)
Solution:

By Newton's second law of motion, $\vec{F}_{net} = m\vec{a}$, hence if the body is moving with constant acceleration then multiple forces may act on body.

(44) **Answer :** (1)
Solution:
$P = 2t + 3t^2$
$\frac{dP}{dt} = 2 + 6t$
F = 2 + 6 × 2 = 14 N

(45) **Answer :** (3)
Solution:

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CHEMISTRY

(46) **Answer :** (3)
Solution:
$\bar{\nu} = \frac{1}{\lambda} = \frac{1}{5000 \times 10^{-10}\text{m}}$
$= 2 \times 10^{6}\ \text{m}^{-1}$
$= 2 \times 10^{4}\ \text{cm}^{-1}$

(47) **Answer :** (3)
Solution:
There is large difference between 3rd and 4th ionisation energy.
$\therefore$ +3 oxidation state will be more stable and the formula of halide is MX_3.

(48) **Answer :** (1)
Hint:
Kinetic energy of electrons increases with the increase of frequency of light.
Solution:
The number of electrons ejected is proportional to the intensity of light.

(49) **Answer :** (2)
Solution:
$E = \frac{hc}{\lambda}$
$\frac{E_1}{E_2} = \frac{\lambda_2}{\lambda_1} = \frac{1}{2}$

(50) **Answer :** (2)
Hint:
$E = \frac{hc}{\lambda}$
Solution:
• Power of bulb = 100 watt = 100 Js^{-1}
• Energy of one photon, $E = \frac{hc}{\lambda}$
$= \frac{6.6 \times 10^{-34} \times 3 \times 10^{8}}{400 \times 10^{-9}}$
$= 4.9 \times 10^{-19}$ J
• Number of photons emitted = $\frac{100\ \text{J}}{4.9 \times 10^{-19}}$
$= 2 \times 10^{20}$

(51) **Answer :** (3)
Solution:
$E_n = -13.6 \left(\frac{Z^2}{n^2}\right)$ eV/atom
$E_3 = -13.6 \left(\frac{2^2}{4^2}\right)$ eV/atom
$= -\frac{13.6}{4} = -3.4$ eV/atom

(52) **Answer :** (1)
Solution:

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The penetrating power of α-particle is the lowest, followed by β-ray (100 times that of α-particle) and γ-rays (1000 times that of α-particle).

(53) **Answer :** (3)
Solution:
Second electron gain ethalpy is endothermic.

(54) **Answer :** (1)
Solution:
Valence shell electronic configuration of Ca is $4s^2$
Quantum numbers : n = 4, l = 0, m = 0, $s = \frac{+1}{2}$ or $\frac{-1}{2}$

(55) **Answer :** (3)
Solution:
Since $\Delta E = \frac{hc}{\lambda}$
∴ Shortest l ⇒ ΔE = Maximum ⇒ last line
So, for Lyman series, $n_2 = \infty \rightarrow n_1 = 1$

$\frac{1}{\lambda} = R_H(1)^2\left(\frac{1}{1^2} - \frac{1}{\infty^2}\right)$

$\frac{1}{x} = R_H$
Also, longest λ ⇒ ΔE = minimum ⇒ first line
So, Balmer series $n_2 = 3 \rightarrow n_1 = 2$

$\frac{1}{\lambda} = R_H(1)^2\left(\frac{1}{2^2} - \frac{1}{3^2}\right)$

$= R_H \times 4 \times \frac{5}{36}$

$\lambda = \frac{9}{5R_H} = \frac{9x}{5}$

(56) **Answer :** (3)
Solution:
For same value of (n + l), higher is the value of n, higher is the energy.

(57) **Answer :** (2)
Solution:
Ga is known as eka-aluminium.

(58) **Answer :** (4)
Solution:
2p subshell can have maximum 6 electrons.

(59) **Answer :** (4)
Solution:
Number of spectral lines $= \frac{\Delta n(\Delta n+1)}{2}$
a. $\Delta n = 4 \Rightarrow \frac{4\times5}{2} = 10$
b. $\Delta n = 1 \Rightarrow \frac{1\times2}{2} = 1$
c. $\Delta n = 2 \Rightarrow \frac{2\times3}{2} = 3$
d. $\Delta n = 5 \Rightarrow \frac{5\times6}{2} = 15$

(60) **Answer :** (4)
Solution:
3rd period contains *s* and *p*-block elements only.

(61) **Answer :** (1)
Solution:
$Na \rightarrow Na^+ + e^-$ E = 5.1 eV
∴ $Na^+ + e^- \rightarrow Na$ E = –5.1 eV

(62) **Answer :** (1)
Hint:
$\frac{1}{\lambda} = R_H \times Z^2\left[\frac{1}{n_1^2} - \frac{1}{n_2^2}\right]$

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Solution:

$\frac{1}{\lambda} \propto Z^2$; higher the value of 'Z', smaller will be the wavelength.

(63) Answer : (3)

Solution:

Orbital angular momentum = $\sqrt{l(l+1)}\hbar$

= $\sqrt{2}\hbar$

(64) Answer : (3)

Solution:

$\because \lambda \ \ \Delta E = R_H\left(\frac{1}{n_i^2} - \frac{1}{n_f^2}\right)$

$\Delta E_1 = R_H\left(\frac{1}{(5)^2} - \frac{1}{(3)^2}\right) = -R_H\frac{16}{225}$

$\Delta E_2 = R_H\left(\frac{1}{(4)^2} - \frac{1}{(3)^2}\right) = -R_H\frac{7}{144}$

$\Delta E_3 = R_H\left(\frac{1}{(3)^2} - \frac{1}{(1)^2}\right) = -R_H\frac{8}{9}$

$\Delta E_4 = R_H\left(\frac{1}{(6)^2} - \frac{1}{(5)^2}\right) = -R_H\frac{11}{900}$

(65) Answer : (2)

Solution:

N : $1s^2\ 2s^2\ 2p^3$

(l) 0 0 1

(66) Answer : (1)

Solution:

Angular momentum (mvr) $= \frac{nh}{2\pi}$

$mvr = \frac{3h}{2\pi}$.

(67) Answer : (1)

Hint:

The species which contain same number of electrons are isoelectronic.

Solution:

Species	Electrons
CO	14
CO^+	13
N_2^-	15
N_2^+	13
C_2^{2-}	14
O_2^{2-}	18
NO	15

(68) Answer : (1)

Solution:

Correct order of energy of 2*s* orbital of the given elements is

$E_{2S}(H) > E_{2S}(Li) > E_{2S}(Na) > E_{2S}(K)$

(69) Answer : (4)

Hint:

Some 2nd period elements have similar properties as of 3rd period elements present diagonally to each other

Solution:

Li is diagonally related to Mg.

Be is diagonally related to Al

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(70) **Answer :** (4)
Solution:

General electronic configuration of Lanthanoids = $6s^2 5d^{0-1} 4f^{1-14}$

(71) **Answer :** (2)
Solution:

n	*l*	Orbital
2	1	2*p*
3	2	3*d*
1	0	1*s*
3	1	3*p*

(72) **Answer :** (4)
Solution:

(A) $[Ne]3s^1$; Na (s-block)

(B) $[Ar]3d^3 4s^2$; V (d-block)

(C) $[Kr]4d^{10} 5s^2 5p^5$; I (p-block)

(D) $[Ar]3d^{10} 4s^1$; Cu (d-block)

(E) $[Rn]5f^0 6d^2 7s^2$; Th (f-block)

Main group elements (A and C only)

(73) **Answer :** (2)
Solution:
Dobereiner's Triads:
(1) Li, Na, K
(2) Ca, Sr, Ba
(3) Cl, Br, I

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(74) **Answer :** (1)
Solution:

Element	Atomic radii/pm
Li	152
Be	111
Na	186
Mg	160

(75) **Answer :** (4)
Solution:

Element	DegH (kJ/mol)
Br	–325
Cl	–349
O	–141
S	–200

(76) **Answer :** (2)
Solution:
Neutral oxides : CO, N_2O, NO

(77) **Answer :** (4)
Solution:
Energy = $N_A \times h\nu$
$= 6.02 \times 10^{23} \times 6.626 \times 10^{-34} \times 3 \times 10^{14}$
$= 119.6 \times 10^3$ J
= 119.6 kJ

(78) **Answer :** (3)
Solution:

F is the most electronegative element with electronic configuration $2s^2 2p^5$

(79) **Answer :** (1)
Solution:

Symbol	Atomic number	IUPAC official name
Unb	102	Nobelium
Unq	104	Rutherfordium
Unh	106	Seaborgium
Unu	101	Mendeleviun

(80) **Answer :** (3)
Solution:
• Canal rays contain positively charged particles which are created by removal of electron from gaseous atom present in discharge tube.

(81) **Answer :** (4)
Hint:
Chromium and copper have exceptional electronic configuration.
Solution:
$_{29}Cu = 1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^{10}\ 4s^1$
$_{24}Cr = 1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^5\ 4s^1$
$_{25}Mn = 1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^5\ 4s^2$
$_{26}Fe = 1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^6\ 4s^2$

Element	Unpaired electron(s)
Cu	1
Mn	5
Cr	6
Fe	4

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(82) **Answer :** (4)
Solution:
$\lambda = \frac{h}{p}$
$\lambda = \frac{6.6\times10^{-34}}{0.01\times10^{-3}\times200}$
$\lambda = 3.3 \times 10^{-31}$ m

(83) **Answer :** (3)
Solution:

Metals	Work function (w_0/eV)
Li	2.42
Na	2.3
K	2.25
Mg	3.7

(84) **Answer :** (4)
Solution:
$\mathrm{h}\nu = \mathrm{h}\nu_0 + \mathrm{K.E.}$
$\mathrm{h}\left(\nu - \nu_0\right) = \mathrm{K.E.}$
$\mathrm{K.E.} = 6.625 \times 10^{-34}\mathrm{J.s}\left(2\times10^{15} - 8\times10^{14}\right)\mathrm{s}^{-1}$
$\mathrm{K.E.} = 6.625 \times 10^{-34}\mathrm{J}\left(12\times10^{14}\right)$
$= 79.5 \times 10^{-20}\mathrm{J} = 7.95 \times 10^{-19}\mathrm{J}$

(85) **Answer :** (3)
Solution:
$\mathrm{n}\lambda = 2\pi\mathrm{r}_1$
or, $\lambda = \frac{2\pi\mathrm{r}_1}{\mathrm{n}}$ .... (i)

$r = a_0 \frac{n^2}{Z}$

$r_1 \left(Li^{2+}\right) = 52.9 \times \frac{1}{3}$ .... (ii)

Putting (ii) in (i)

$\lambda = \frac{2\pi \times 52.9}{3}$

$\lambda = 35.3\pi \text{ pm}$

(86) **Answer :** (1)
Solution:
Only Lyman series lies in Ultraviolet Region.

(87) **Answer :** (2)
Solution:
0.04 gram atom of magnesium require = y kJ
1 mole of magnesium require = $\frac{y}{0.04}$ kJ= 25y kJ
10 atoms will required = $\frac{25y \times 10}{N_A} = \frac{250\,y}{N_A}$

(88) **Answer :** (1)
Solution:
s-orbital is symmetrically distributed in all direction.

(89) **Answer :** (2)
Solution:
A (Z = 39) is yttrium present in *d*-block (3rd group).

(90) **Answer :** (4)
Solution:
$(r_2)_{He^+} = 0.529 \times \left(\frac{2^2}{2}\right)$ Å
= 1.058 Å

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BOTANY

(91) **Answer :** (3)
Solution:
Cell growth results in disturbing the ratio between the nucleus and the cytoplasm. Mitotic divisions in the meristematic tissues – the apical and the lateral cambium, result in a continuous growth of plants throughout their life.

(92) **Answer :** (3)
Solution:
The chromosomes become compact→the chromosomes align at the equator→four haploid daughter cells formation

(93) **Answer :** (1)
Solution:
Human cells in culture divide once in approximately every 24 hours.

(94) **Answer :** (2)
Solution:
In cytokinesis of animal cells, furrow starts at the periphery and then moves inwards centripetally, dividing the cell into two halves.

(95) **Answer :** (3)
Solution:
Quiescent stage of cell cycle is G_0-phase. Cells in G_0 stage remain metabolically active but no longer proliferate unless called on to do so depending on the requirement of the organism.

(96) **Answer :** (3)
Solution:
The key events of telophase are -chromosomes cluster at opposite spindle poles and their identity is lost as discrete elements. Nuclear envelope develops around the chromosome clusters at each pole forming two daughter nuclei.

(97) **Answer :** (2)

Solution:
Mitosis is a type of cell division that results in two daughter cells, each having the same number of chromosomes and the same amount of DNA as the parent cell. This process ensures that the genetic material is equally distributed, maintaining genetic consistency across cells.

(98) Answer : (2)
Solution:
During telophase stage, nucleolus, endoplasmic reticulum, golgi bodies and other organelle reappear in the daughter cell.

(99) Answer : (2)
Solution:
1 bivalent = 1 pair of chromosomes, i.e., 2 chromosomes. Therefore, 15 bivalents = 30 chromosomes each having sister chromatids.

(100) Answer : (3)
Solution:
If karyokinesis is not followed by cytokinesis as a result of which multinucleate condition arises leading to the formation of syncytium.

(101) Answer : (3)
Solution:
Number of meiosis for the foarmation of n pollen grains = $\frac{n}{4} = \frac{100}{4}$ = 25
Each microspore mother cell undergoes one meiotic division to produce 4 pollen grains. Therefore, to form 100 pollen grains, we need 100/4 = 25 meiotic divisions.

(102) Answer : (2)
Solution:
The diplotene stage is initiated by the dissolution of the synaptonemal complex. During this stage, the homologous chromosomes begin to separate, except at the sites of crossovers, which are visible as chiasmata.

(103) Answer : (2)
Solution:
In G_1 phase, most of the cell organell duplicates. in M phase or mitotic phase, actual cell division will take place. In S phase centriole will duplicate in the cytoplasm and DNA in nuclues. G_0 phase is also known quiescent phase.

(104) Answer : (3)
Hint:
During the second phase of karyokinesis, mitotic spindle formation is completed.
Solution:
The microtubules of the spindle fibres attach to the kinetochore during the metaphase.

(105) Answer : (4)
Solution:
Cytokinesis marks the end of the cell division.

(106) Answer : (2)
Solution:
The best stage to study the shape of chromosomes is anaphase.

(107) Answer : (3)
Solution:
Meiosis is the mechanism in which conservation of specific chromosome number of each species is achieved across generation in sexually reproducing organisms.

(108) Answer : (3)
Solution:
In male honey bee gamete formation occurs by mitosis as this organism is haploid.

(109) Answer : (3)
Hint:
Meristematic cells undergo mitosis.
Solution:
Bivalents or homologous chromosome pairs are formed in zygotene stage of meiosis. Meristems being somatic cells do not perform meiosis.

(110) Answer : (2)
Hint:
Bringing of the chromosomes on the equator of spindle is called congression.
Solution:

The phenomena of congression of chromosomes is observed in the metaphase stage.

(111) Answer : (1)
Hint:
In meiosis I, the enzyme recombinase is activated during crossing over.
Solution:
Crossing over occurs during pachytene stage of prophase I.

(112) Answer : (4)
Solution:
During interkinesis stage, RNA and proteins are synthesized.

(113) Answer : (2)
Hint:
Crossing over occurs during meiosis.
Solution:
Meiosis produces gametes and the process of gamete formation is called gametogenesis.

(114) Answer : (3)
Solution:
Archaebacteria differ from other bacteria in having a different cell wall structure and this feature is responsible for their survival in extreme conditions.

(115) Answer : (2)
Solution:
Taxonomy is a process of classification that is based on the characteristics of all living organisms that can be classified into different taxa.
Systematics takes into account evolutionary relationships between organisms.
Classification is a process by which anything is grouped into convenient categories based on some easily observable characters.
Taxonomic categories are distinct biological entities and not merely morphological aggregates.

(116) Answer : (1)
Solution:
Musca is genus of housefly.

(117) Answer : (4)
Solution:
Modern taxonomic studies are based on external and internal structure of organisms along with cell structure, development process and ecological information of organisms, whereas the classical taxonomy is based on observable morphological characters.

(118) Answer : (4)
Solution:

Family	**Order**
Potato – Solanaceae	Polymoniales
Mango – Anacardiaceae	Sapindales

Both polymoniales and sapindales are included in class dicotyledonae.

(119) Answer : (2)
Hint:
Lion and leopard are two different species but both belong to the genus, *Panthera*.
Solution:
Plant family Convolvulaceae is included in the order Polymoniales, mainly on the basis of floral characters.

(120) Answer : (4)
Solution:
Correct scientific name according to binomial nomenclature is *Triticum aestivum* Linn.

(121) Answer : (2)
Solution:
First word in a biological name represents the genus while the second component denotes specific epithet.

(122) Answer : (2)
Solution:
Tiger, mammals, and animals represent different taxa at different levels. Tiger is a species, mammals are a class, and animals belong to the kingdom.

(123) Answer : (1)
Solution:

Taxonomic categories showing hierarchial arrangement in ascending order :
Species → Genus → Family → Order → Class → Phylum / Division → Kingdom.

(124) Answer : (4)
Solution:
Mango belongs to the order Sapindales.

(125) Answer : (2)
Hint:
The scientific name of brinjal is *Solanum melongena*.
Solution:
The scientific name of potato is *Solanum tuberosum*.
Thus, only specific epithets of brinjal and potato are different.

(126) Answer : (1)
Solution:
The first word in a biological name represents the genus while the second component denotes the specific epithet.

(127) Answer : (3)
Solution:
Datura belongs to the family Solanaceae.

(128) Answer : (3)
Solution:
The number of species that are known and described range between 1.7-1.8 million.

(129) Answer : (3)
Solution:
The main criteria for classification used by him include cell structure, body organisation, mode of nutrition, reproduction and phylogenetic relationships.

(130) Answer : (3)
Solution:
The bacteria that can live in extreme saline conditions are Halophiles.

(131) Answer : (3)
Hint:
First scientific attempt for classification was performed by Aristotle.
Solution:
Aristotle classified organisms on the basis of simple morphological features.

(132) Answer : (3)
Hint:
Both algae and fungi include eukaryotic organisms.
Solution:
Algae are photosynthetic organism and fungi are non-photosynthetic organism.

(133) Answer : (2)
Solution:
Correct sequence of cell cycle is $G_1 \rightarrow S \rightarrow G_2 \rightarrow M$

(134) Answer : (4)
Solution:
The bacteria found in the gut of ruminant animals that produce methane from dung are methanogens. Methanogens are responsible for the production of methane, a component of biogas, in the digestive systems of ruminants.

(135) Answer : (3)
Solution:
The three domain system proposes two domains for prokaryotes and one for eukaryotes.

ZOOLOGY

(136) Answer : (3)
Solution:
Conducting part of respiratory system extends from external nostrils upto terminal bronchioles.

(137) Answer : (2)

Solution:
Prosthetic groups are organic compounds and are distinguished from other co-factors in that they are tightly bound to the apoenzyme. For example, in peroxidase and catalase, which catalyse the breakdown of hydrogen peroxide to water and oxygen, haem is the prosthetic group and it is a part of the active site of the enzyme.
• Zinc is a co-factor for the enzymes, carboxypeptidase and carbonic anhydrase.

(138) Answer : (2)
Hint:
Neutral, optically active amino acid.
Solution:
R group in glycine is hydrogen, while in serine, hydroxy methyl group is the R group.

(139) Answer : (2)
Solution:
Trachea, primary, secondary, tertiary and initial bronchioles are supported by incomplete cartilaginous rings.

(140) Answer : (3)
Solution:
Volume of air inspired or expired during a normal respiration is called tidal volume (TV = 500 mL).

(141) Answer : (3)
Solution:
Antibodies are categorised as proteins.

(142) Answer : (2)
Hint:
Leucine belongs to this category.
Solution:
There are 20 types of amino acids.
The amino acids which can't be synthesized by our body have to be supplied through our diet. These are called essential amino acids.
The amino acids which our body can make are non-essential amino acids.

(143) Answer : (1)
Solution:
Trachea at the level of 5th thoracic vertebra divides into right and left primary bronchi.

(144) Answer : (2)
Hint:
An enzyme present in plants.
Solution:
Chitin is present in the exoskeleton of arthropods. Collagen is the most abundant protein in the animal world, while Ribulose Bisphosphate Carboxylase-Oxygenase (RuBisCO) is the most abundant protein in whole of the biosphere.

(145) Answer : (1)
Solution:
Ribozyme is a nucleic acid (RNA) that acts as an enzyme and catalyzes a chemical reaction.

(146) Answer : (2)
Hint:
R group in glycine is H.
Solution:

Beta-pleated sheet	–	Secondary structure of protein
Glycosidic bond	–	Exists between two simple sugars in oligosaccharide
Disulphide bond	–	Present in tertiary and quaternary structure of proteins
Glycine	–	Simplest amino acid

(147) Answer : (2)
Solution:
A triglyceride consists of 3 molecules of fatty acids attached to a glycerol molecule.

(148) Answer : (3)
Solution:
Guanine pairs with cytosine and adenine pairs with thymine in a DNA molecule.

(149) Answer : (1)
Solution:
Total volume of air a person can expire after a normal inspiration is called expiratory capacity. This includes TV and ERV.

(150) Answer : (1)
Solution:
Co-enzymes NAD and NADP contain the vitamin niacin.

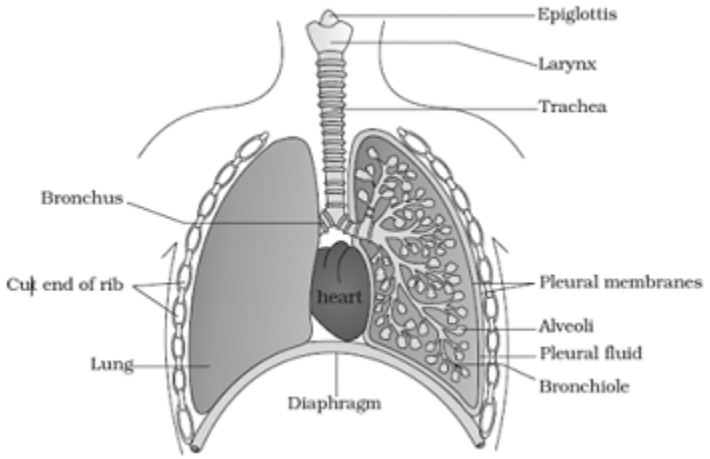
(151) Answer : (3)
Solution:
Activity cannot be restored in denatured enzymes.

(152) Answer : (2)
Solution:
Tyrosine, phenylalanine and tryptophan are examples of aromatic amino acids.

(153) Answer : (2)
Solution:
Each lung is enclosed in two membranes called pleurae. The outer covering is adhered to chest wall and diaphragm and is called parietal pleura. The inner covering membrane, which closely covers the lung, is called visceral pleura.

(154) Answer : (1)
Solution:
Competitive inhibition could be reversed by increasing the concentration of substrate.

(155) Answer : (3)
Solution:



(156) Answer : (3)
Hint:
Free amino group is present in first amino acid.
Solution:
In a long peptide chain, the first amino acid contains free amino group, hence it is also called the N-terminal and the last amino acid contains free carboxyl group, hence it is also called C-terminal amino acid. The first amino acid is present at left end, called N-terminal and the last amino acid is present at right end, called C-terminal.

(157) Answer : (4)
Solution:
The activity of an enzyme can be affected by a change in the conditions which can alter the tertiary structure of the protein. These include temperature and pH. Change in substrate concentration or binding of specific chemicals can also regulate the activity of the enzyme.
Activation energy does not affect the catalytic activity of an enzyme.

(158) Answer : (1)
Solution:
Lung ventilation movements are due to contraction and relaxation of the muscles involved in breathing.

(159) Answer : (2)
Solution:
Inspiration is defined as a process by which fresh air enters into the lungs. Diaphragm and external inter-costal muscles play an important role in inspiration. The pressure within the lungs is made lesser than the atmospheric pressure so that air can flow into lungs by contraction of muscles involved.

(160) Answer : (1)
Solution:

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During normal inspiration, the external intercostal muscles contract that lifts the ribs and sternum up and outward causing an increase in the volume of thoracic cavity in the dorso-ventral axis. Intra-pulmonary pressure is always negative during normal inspiration. The diaphragm flattens during normal inspiration.

(161) Answer : (4)
Solution:
Exchange part is the site of actual diffusion of O_2 and CO_2 between blood and atmospheric air.

(162) Answer : (1)
Solution:
Uridylic acid is a nucleotide.

(163) Answer : (2)
Solution:
FRC = 2500 mL, VC = 3500 – 4500 mL;
IC = 3000 – 3500 mL
RMV = 6000 – 8000 mL

(164) Answer : (4)
Solution:
TV is approx 500 mL.
Residual volume : This averages 1100 mL to 1200 mL
Inspiratory reserve volume : This averages 2500 mL to 3000 mL
Vital capacity : This averages 4000 mL to 4600 mL

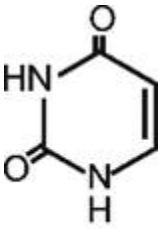
(165) Answer : (4)
Hint:
The animal contains haemolymph
Solution:
Cockroaches do not have capillary beds within their tracheal system.
For earthworms, as well as some amphibians (frogs), the skin serve as the respiratory organ.
A dense network of capillaries just below the skin facilitates the exchange of gases between the circulatory system and the environment.
Gills of fishes also contain a dense network of capillary beds.

(166) Answer : (3)
Solution:
The two strands of DNA are held together by hydrogen bonds.
Glycosidic bonds connect the nitrogenous base to the deoxyribose sugar within a single nucleotide.

- Peptide bonds link amino acids together to form proteins, not nucleic acids.
-
- Coordination bonds are not involved in holding the primary or secondary structure of the DNA strands together.

(167) Answer : (1)
Hint:
Hemoglobin is tetrameric compound.
Solution:
Adult human haemoglobin represents quaternary structure of a protein and it contains two subunits of α type polypeptide chains and two subunits of β type polypeptide chains. So, it is a tetrameric protein molecule with iron in Fe^{+2} state.

(168) Answer : (2)
Hint:
Presence of 2 H-bonds
Solution:



Uracil

Three hydrogen bonds are present between guanine and cytosine, while two hydrogen bonds are present between adenine and thymine. Arachidonic acid is an unsaturated fatty acid.

(169) Answer : (3)
Solution:

Co-enzymes are organic compounds. Their association with the apoenzyme is only transient, usually occurring during the course of catalysis.

(170) Answer : (3)
Hint:
More than one factors influence mechanism of breathing.
Solution:
Mechanism of breathing vary among different groups of animals depending mainly on their habitats and levels of organisation.

(171) Answer : (3)
Solution:
The lungs are situated in the thoracic chamber which is anatomically an air-tight chamber.

(172) Answer : (1)
Solution:
Breathing involves exchange of gases between lungs and atmosphere.

(173) Answer : (4)
Solution:
A part of the pharynx provides passage to both air and food.

(174) Answer : (1)
Solution:
In an enzyme-catalysed reaction, rate doubles or decreases by half for every 10°C change in either direction.

(175) Answer : (4)
Solution:
Upon hydrolysis, a nucleoside yields nitrogenous base and pentose sugar.

(176) Answer : (2)
Solution:
Chitin is a homopolysaccharide.
Cellulose is the structural polysaccharide of the plant cell wall.
Lecithin is a phospholipid.
Insulin and tryspin are proteins.

(177) Answer : (3)
Solution:
Lungs are present in an air tight-chamber known as thoracic cavity in humans.
Such an arrangement is essential for breathing, as we cannot directly alter the pulmonary volume.

(178) Answer : (2)
Solution:
Amino acids are organic compounds containing an amino group and an acidic group as substituents on the same carbon i.e., the α-carbon.

(179) Answer : (4)
Solution:
Except for FRC, TLC and RV, all other volume/capacities can be measured by using a simple spirometer.

(180) Answer : (2)
Solution:
Larynx is a cartilaginous box which helps in sound production and hence called the sound box.

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